

A World Model, From Scratch

A functioning world model built from first principles in PyTorch — trained on a laptop (Apple M4 Pro / MPS). It learns a 2D physics world from raw pixels, hallucinates that world forward in imagination, and trains a controller entirely inside its own dream. Reproduces Ha & Schmidhuber's V + M + C decomposition (2018) — no gym, no RL library, no pretrained weights. End-to-end in ~13 minutes.

What Is a World Model?

A system that builds an internal representation of an environment and predicts how it changes in response to actions. The idea predates deep learning: Kenneth Craik argued in 1943 that the mind runs a "**small-scale model**" of reality; RL formalized it as a POMDP — an agent acts, a hidden state changes, and it observes only partially.

1

Genie 3

Real-time playable worlds

2

DreamerV3

Nature, 2025

3

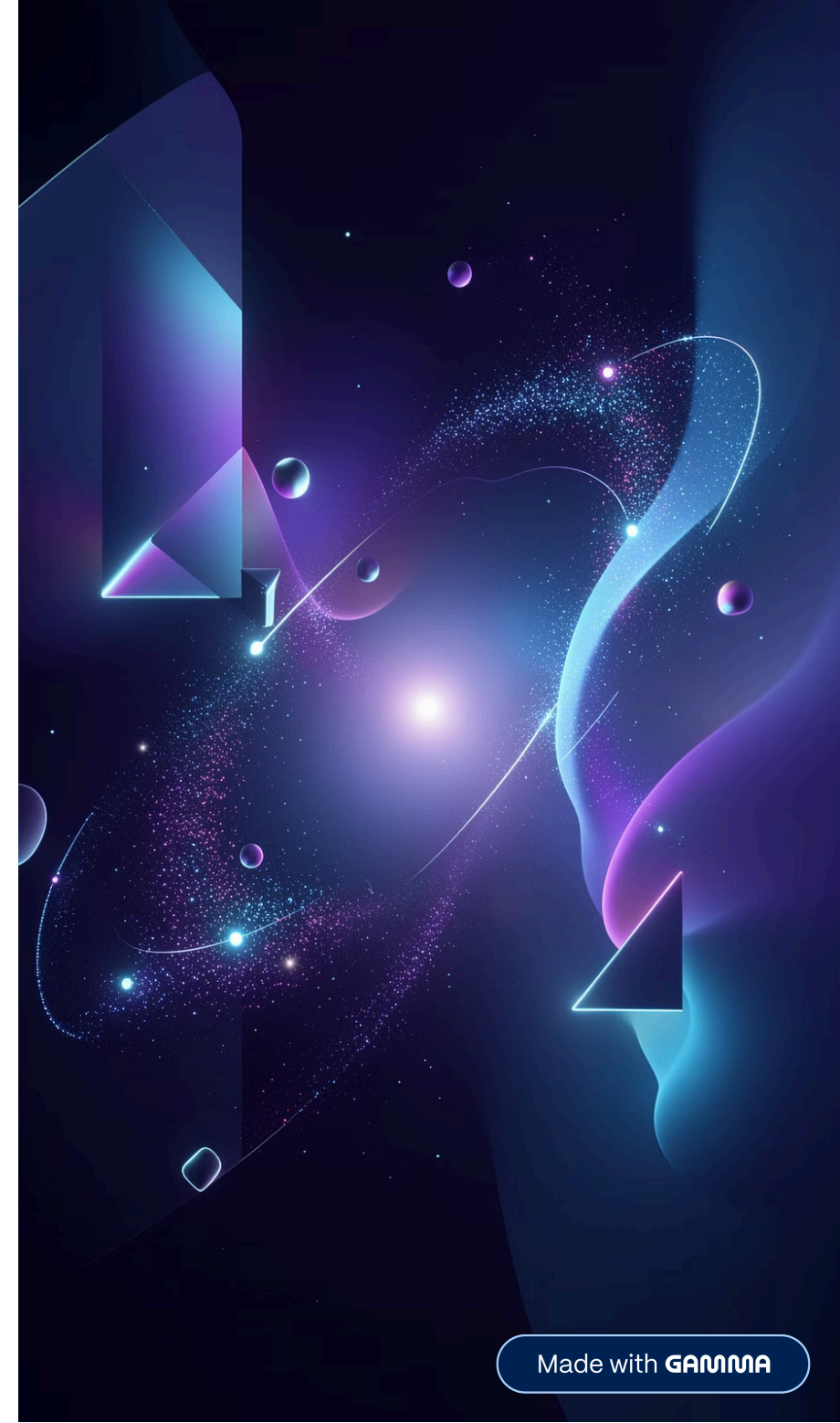
Sora & Cosmos

OpenAI + NVIDIA

4

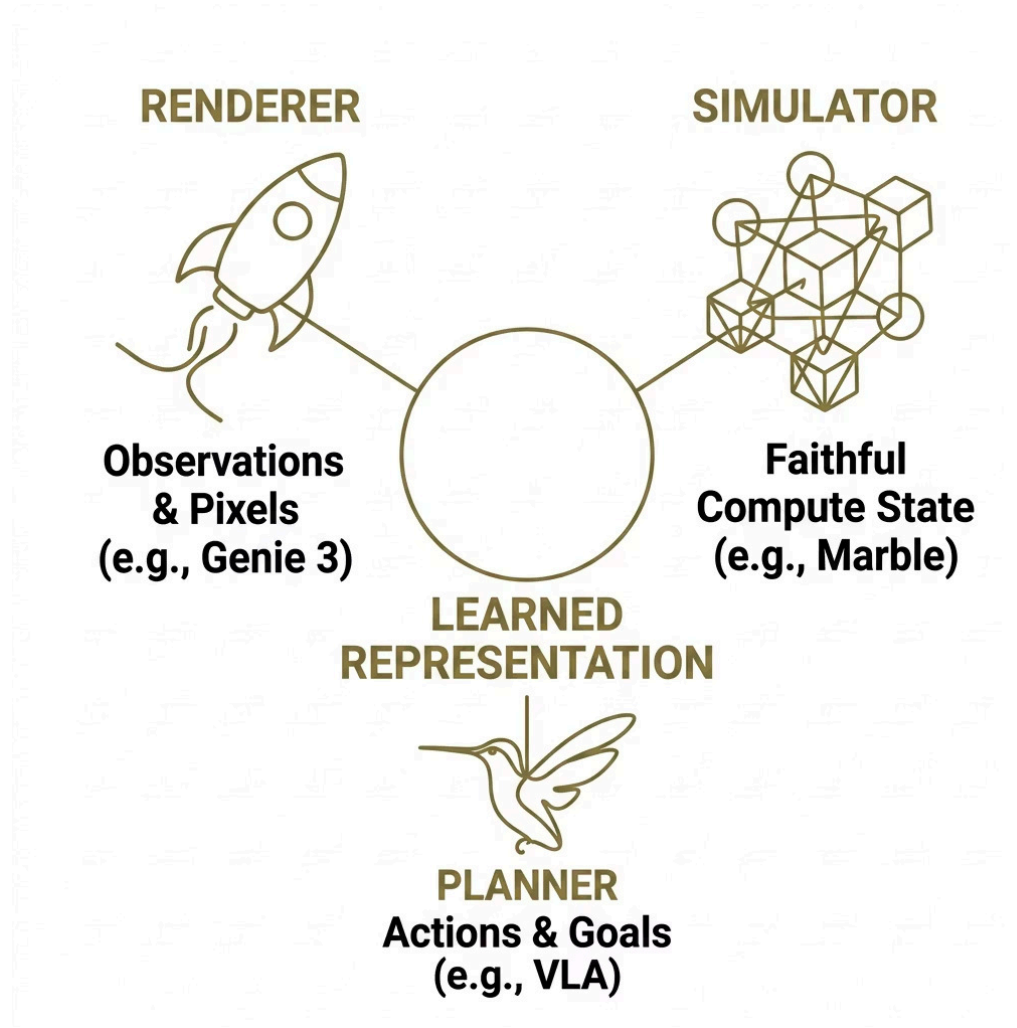
World Labs

Fei-Fei Li, 2024–26



Fei-Fei Li's Functional Taxonomy

Classifies world models by what they **output** — June 2026. Her thesis: these are *"three projections of a single underlying understanding."*



Renderer

Observations / pixels for human eyes — e.g. **Genie 3**

Simulator

Faithful state you can compute on — e.g. **World Labs' Marble**

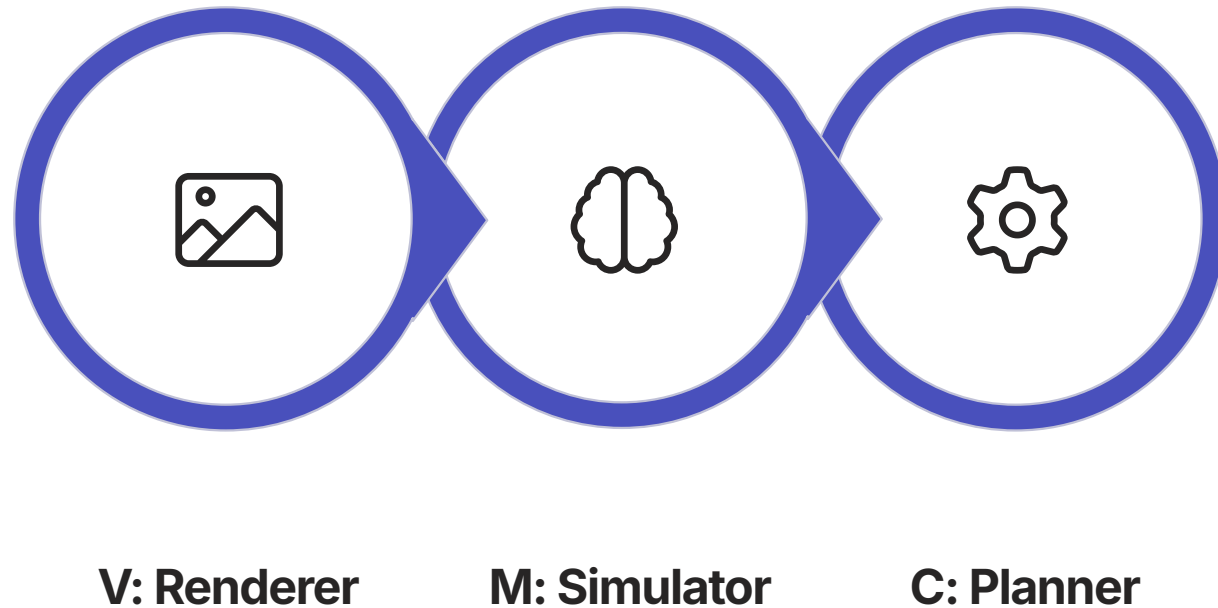
Planner

Actions toward a goal — e.g. **Vision-Language-Action models**

- ❑ This project is a miniature of exactly that — one learned representation that is rendered, simulated, and planned over.

The Architecture: **V + M + C**

Heavy world model, light policy — the architectural DNA since Schmidhuber, 1990.



Each component has a distinct role: **V** perceives, **M** imagines, and **C** decides — mirroring Fei-Fei Li's renderer/simulator/planner taxonomy at minimal scale.

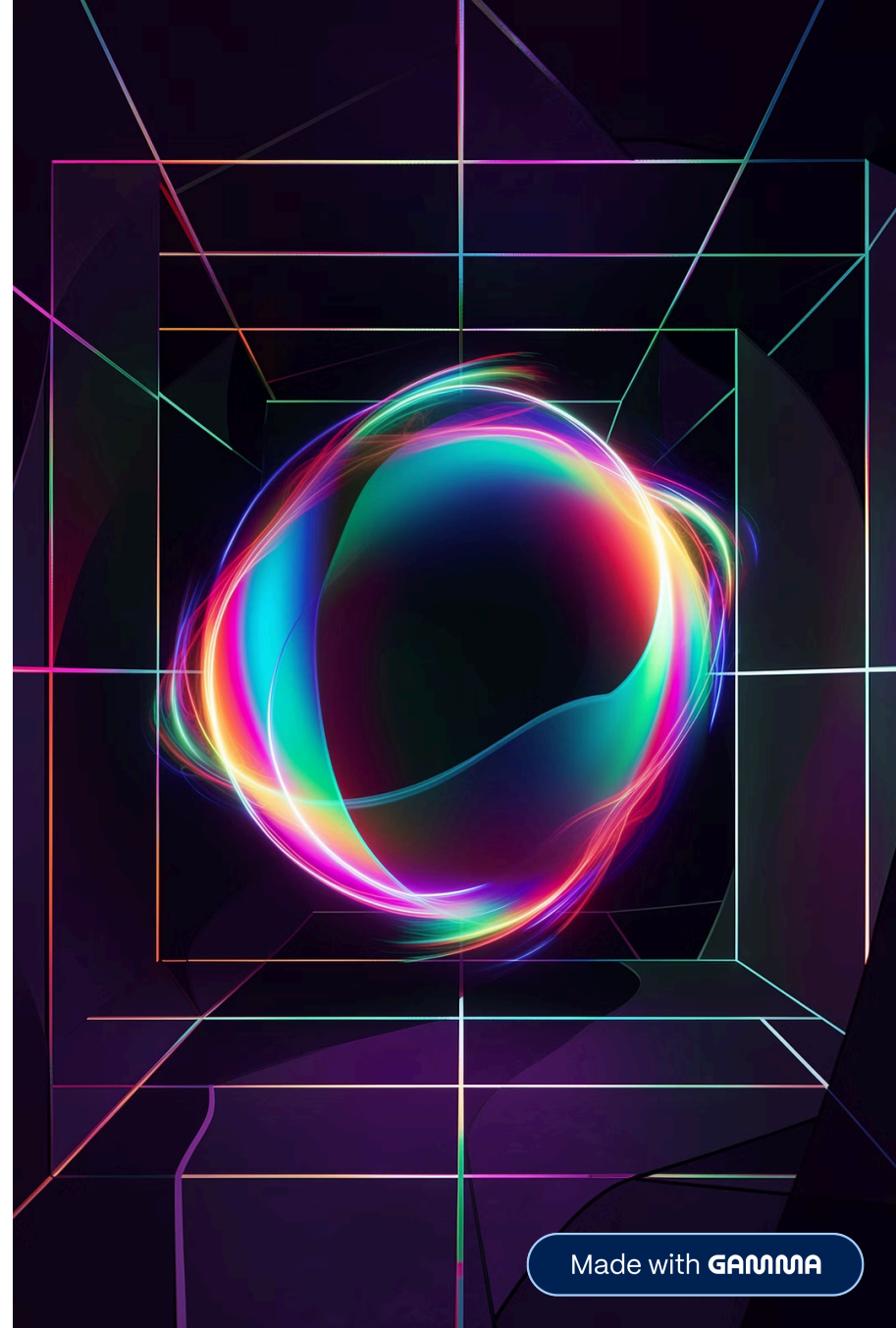
The Environment, Also From Scratch

Physics

Thrusted ball in a walled 2D arena — friction, speed cap, elastic wall bounces, and a static goal square. Pure NumPy — no gym, no physics engine.

Dataset

320 episodes × 150 steps = 48,000 frames collected in 4 seconds under a random policy. Simple enough that dream quality is judgeable by eye.



V — The VAE (Renderer)

Four strided convolutions down to a **32-D Gaussian latent**, four transposed convolutions back up.

The Surprise: Foreground-Weighted Loss

Plain pixel-MSE **erased the tiny moving ball** — the dark background dominated the loss. Fix: upweight non-background pixels **30x**.

- ✓ The payoff: a linear probe decodes ball (x, y) position with **$R^2 = 0.97$** and ~3-pixel error. Position was never a label — the VAE discovered the world's true state variable on its own.

Encoder

4× strided conv → 32-D
 μ, σ

Latent

Gaussian
reparameterization

Decoder

4× transposed conv → 64×64

M — The MDN-RNN (Simulator)

An LSTM that takes $[z, \text{action}]$ and predicts a **mixture of 5 Gaussians** over the next latent. Modeling a **distribution** rather than a point is what enables sampling plausible futures — i.e., the ability to *dream*.

01

Encode

One real frame $\rightarrow$ latent z_0 via V

02

Rollout

Hallucinate every following frame from actions alone via M — no real observations

03

Decode

Latents $\rightarrow$ pixels via V decoder, only for display

The Dream Works — With an Honest Caveat

Real Frame	VAE Reconstruction
Ground truth from environment	One-step encode → decode
Open-Loop Dream	
160-frame hallucination from single seed	

Side-by-side, the dream tracks the dynamics: the ball keeps bouncing and the goal stays in place across a fully hallucinated 160-frame rollout.



Long-horizon drift: Small per-step errors compound over time — the same failure mode flagged across Sora, DreamerV3, and latent video models. Dynamics stay *qualitatively* correct; quantitative precision degrades.

C — Trained Entirely Inside the Dream

A from-scratch **Cross-Entropy Method** evolves a **1,445-parameter linear policy**. Fitness measured only on imagined rollouts; reward derived from the latent→position probe. The controller *never touches the real environment* during training.

39%

Dream-trained policy

Goal reach rate vs. 5% for random
— **~8× gain**

0.136

Mean closest distance

vs. 0.358 for random — more than
halved over 100 unseen pairs

12%

Over-trained policy

Model exploitation: fixed
deterministic dreams → learned the
model's flaws, not reality

Key lesson: Resampling dream scenarios + validation-in-
imagination fixed exploitation. Diversity in imagination is as critical
as quality.

Where This Sits in the Landscape



Genie 3

Interactive playable worlds from video



DreamerV3

150+ tasks; mines Minecraft diamonds



NVIDIA Cosmos

Physical AI and robotics simulation



LeCun's JEPA

Predict in representation space, not pixels

From a 34-agent, 61-source research sweep, this project is **the smallest honest version of those ideas** — V + M + C, from scratch, on a laptop.



Reproduce everything with one command:

```
bash run_all.sh
```

~13 minutes. Apple M4 Pro / MPS. No pretrained weights. No RL library. No gym.